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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING
SLOT : A

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I P.Mokshith Reddy/192425149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mokshith

Name of the student:

mokshith reddy

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05

Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

MLA04 – CO3 Comparative Analysis Answers

The uploaded assessment contains **5 questions for 25 marks**, covering CNN architecture, filters, parameter sharing, AlexNet/ResNet, and regularization.

Q1. CNN vs Fully Connected Neural Network

Parameter	CNN	Fully Connected Neural Network
Architecture	Uses convolution, pooling and fully connected layers	Mainly uses fully connected layers
Motivation	Designed specifically for spatial data such as images	General-purpose feature learning
Spatial feature extraction	Preserves spatial relationships	Spatial relationships are largely lost after flattening
Parameters	Fewer due to local connections and parameter sharing	Very large number of parameters
Computation	More efficient for images	Computationally expensive for high-resolution images
Image handling	Excellent	Less suitable
Feature learning	Automatically learns edges, textures and shapes	Learns features but without explicit spatial structure

Role of convolution: Convolutional layers apply learnable filters to detect local patterns.

Role of filters: Filters detect features such as edges, corners, textures and shapes.

Role of pooling: Pooling reduces feature-map dimensions, computation and sensitivity to small positional changes.

Conclusion: CNNs are more suitable for X-ray classification because they preserve spatial information, require fewer parameters and automatically learn hierarchical image features. This makes them more efficient and effective than traditional fully connected networks.

Q2. Different CNN Layers and Filters

Layer	Main Function	Spatial Information	Computational Requirement
Convolution	Extracts features	Preserves spatial structure	Moderate
Pooling	Reduces dimensions	Partially reduces spatial detail	Low
Activation	Introduces non-linearity	Maintains feature-map structure	Low
Fully Connected	Performs classification	Uses learned features	High when input is large

Early vs Deep Filters

Early-layer filters:

Detect simple features.

Learn edges, lines and basic textures.

Usually operate directly on the input image.

Deeper-layer filters:

Combine simple features into complex patterns.

Detect shapes, object parts and high-level structures.

Provide more meaningful representations for classification

Conclusion

The layers work together hierarchically. Convolution extracts features, activation introduces non-linearity, pooling reduces unnecessary spatial information, and fully connected layers perform classification. Therefore, a combination of these layers can effectively identify industrial defects at different levels of complexity.

Q3. Parameter Sharing vs Independent Weights

Parameter	Parameter Sharing in CNN	Independent Weights in FCN
Weights	Same filter reused across locations	Different weights for different connections
Number of parameters	Much lower	Very high
Memory	Lower	Higher
Computation	Efficient for images	Expensive for high-resolution images
Feature detection	Detects the same feature at different locations	Location-specific
Translation property	Provides useful translation-related robustness	Limited
Learning efficiency	High	Lower for image data

Importance of Parameter Sharing

In a CNN, one filter is applied across different positions of an image. For example, an edge-detection filter can detect an edge whether it occurs on the left, center or right side.

This greatly reduces the number of trainable parameters and memory requirements. It also allows the network to learn reusable visual features efficiently.

Conclusion

Parameter sharing is essential for large-scale image processing because thousands of high-resolution images can be processed with considerably fewer parameters than a fully connected architecture.

Q4. AlexNet vs ResNet

Parameter	AlexNet	ResNet
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Network depth	Relatively shallow	Much deeper
Layer organization	Conventional sequential CNN	Uses residual blocks
Filters	Convolutional filters with progressively higher-level features	Convolutional filters within residual blocks
Parameters	Relatively large	Depends on variant; efficient relative to depth
Feature extraction	Good	Excellent for complex features
Training difficulty	Easier than very deep networks	Designed to make deep networks easier to train
Vanishing gradient	More manageable due to lower depth	Skip connections strongly help gradient flow
Computational cost	Lower than many deep ResNets	Generally higher
Performance	Major improvement for its time	Usually superior for complex recognition tasks

AlexNet

AlexNet demonstrated the effectiveness of deep CNNs for large-scale image classification. It uses convolutional layers followed by fully connected layers to learn hierarchical image features.

ResNet

ResNet introduces **residual/skip connections**, allowing information and gradients to bypass some layers. This makes it possible to train much deeper networks.

Recommendation

For a complex image-recognition application, **ResNet is the more suitable choice** because its deeper architecture can learn richer representations and residual connections help overcome degradation and vanishing-gradient difficulties.

Q5. Regularized CNN vs Non-Regularized CNN

Parameter	Non-Regularized CNN	Regularized CNN	
Overfitting	High	Reduced	
Generalization	Poor on unseen data	Better	
Training accuracy	Can become very high	Usually controlled	
Training stability	May be unstable/generalize poorly	Batch normalization can improve stability	
Computational requirement	Lower	Slightly higher	

Real-world performance	Often poor if overfitted	Generally better	
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1. Dropout

Dropout randomly deactivates a portion of neurons during training. This prevents the network from depending too heavily on particular neurons and helps reduce overfitting.

2. Data Augmentation

Data augmentation creates modified training images through transformations such as rotation, flipping, cropping or scaling. It increases the effective diversity of the training dataset.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training. It can improve training stability and often allows more efficient optimization.

Recommended Approach

For plant-disease classification, the CNN should use **data augmentation + dropout + batch normalization**.

- Data augmentation** improves dataset diversity.
- Dropout** reduces over-dependence on specific neurons
- Batch normalization** improves training stability

Final Conclusion

Since the initial model has high training accuracy but poor performance on unseen images, it is clearly suffering from **overfitting**. A regularized CNN is therefore more appropriate for reliable real-world plant-disease classification.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand